Note: $E_\pm(\vec{k}) = E_\pm(-\vec{k})$ due to time-reversal $(t \to -t)$ symmetry $\vec{k} \to -\vec{k}$

$\left(\vec{k} = \vec{p}/\hbar \,,\ \vec{p} \sim \frac{d\vec{r}}{dt} \longrightarrow -\vec{p}\right)$ when $t \to -t$

$\Rightarrow$ if $E_\pm(\vec{k} = \vec{K}) = 0$

we immediately know that $E_\pm(\underbrace{-\vec{K}}_{\vec{K}'}) = 0$ too.

Since $\vec{K}$ and $\vec{K}' = -\vec{K}$ are not related by the reciprocal lattice vector, they represent ~~this~~ two-fold degeneracy also known as <u>valley</u> degeneracy.

Eigenfunctions: $\mathcal{H}(\vec{k})\,\psi_\lambda(\vec{k}) = E_\lambda(\vec{k}) \quad (\lambda = \pm 1)$

Let $\psi_\lambda(\vec{k}) = \begin{pmatrix} a_k^\lambda \\ b_k^\lambda \end{pmatrix}$. Note that a_k^λ (b_k^λ) represent component of wavefunction of A(B) sublattice.

$$\begin{pmatrix} 0 & g(k) \\ g^*(k) & 0 \end{pmatrix} \begin{pmatrix} a_k^\lambda \\ b_k^\lambda \end{pmatrix} = E_\lambda(k) \begin{pmatrix} a_k^\lambda \\ b_k^\lambda \end{pmatrix} = \underbrace{\lambda\,|g(\vec{k})|}_{E_\pm} \begin{pmatrix} a_k^\lambda \\ b_k^\lambda \end{pmatrix}$$

$$\Rightarrow \quad g(k)\, b_k^\lambda = \lambda\, |g(k)|\, a_k^\lambda$$

$$\Rightarrow \quad a_k^\lambda = \lambda \frac{g(k)}{|g(k)|}\, b_k^\lambda = \lambda\, e^{-i\varphi_k}\, b_k^\lambda$$

where $e^{-i\varphi_k} \equiv \dfrac{g(k)}{|g(k)|}$ and $g(\vec{k}) = (-t)\sum_{j=1}^{3} e^{i\vec{k}\cdot\vec{e}_j} =$

$$= (-t)\, 2\cos\left(\frac{k_x a}{2}\right) e^{-i\frac{k_y a}{2\sqrt{3}}} + e^{i\frac{k_y a}{\sqrt{3}}}$$

Normalization: $\psi_\lambda^+(\vec{k})\,\psi_\lambda(\vec{k}) = \left(a_{\lambda k}^*,\, b_{\lambda k}^*\right) \cdot \begin{pmatrix} a_{\lambda k} \\ b_{\lambda k} \end{pmatrix} = 1$

$$\Rightarrow |a_{\lambda k}|^2 + |b_{\lambda k}|^2 = 1$$

$$\Rightarrow |a_{\lambda k}| = \frac{1}{\sqrt{2}}$$

$$\psi_\lambda(\vec{k}) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \lambda e^{+i\varphi_k} \end{pmatrix} \quad \text{or} \quad \frac{1}{\sqrt{2}} \begin{pmatrix} e^{-i\varphi_k} \\ \lambda \end{pmatrix}$$

(differ by phase $e^{-i\varphi_k}$)

Low-energy continuum limit:

lowest energy states are located in the vicinity of $\vec{K}$ and $\vec{K}' = -\vec{K}$ points $\Rightarrow$ makes sense to derive effective low-energy Hamiltonian near $\pm\vec{K}$.

let $\vec{k} = \pm\vec{K} + \vec{q}$, $\vec{K} = \left(\frac{4\pi}{3a}, 0\right)$

↳ momentum measured from vector $(\pm\vec{K})$.

$$g(\vec{k}) = g(\pm\vec{K} + \vec{q}) = \sum_{j=1}^{3} e^{i(\pm\vec{K}+\vec{q})\cdot\vec{e}_j} = \text{(see p. 77)}$$

$$= e^{i\vec{q}\cdot\vec{e}_1} + e^{\pm i\vec{K}\cdot\vec{e}_2} e^{i\vec{q}\cdot\vec{e}_2} + e^{\pm i\vec{K}\cdot\vec{e}_3} e^{i\vec{q}\cdot\vec{e}_3}$$

$$\left(\vec{e}_1 = \frac{a}{\sqrt{3}}\vec{e}_y \Rightarrow \vec{K}\cdot\vec{e}_1 = 0.\right)$$

$$\vec{e}_2 = -\frac{a}{2}\left(\vec{e}_x + \frac{\vec{e}_y}{\sqrt{3}}\right) \Rightarrow \vec{K}\cdot\vec{e}_2 = \frac{4\pi}{3a}\left(-\frac{a}{2}\right) = -\frac{2\pi}{3}$$

$$\vec{e}_3 = \frac{a}{2}\left(\vec{e}_x - \frac{\vec{e}_y}{\sqrt{3}}\right) \qquad \vec{K}\cdot\vec{e}_3 = \frac{2\pi}{3}$$

$$g(\pm\vec{K}+\vec{q}) = e^{i\frac{q_y a}{\sqrt{3}}} + e^{\mp i\frac{2\pi}{3}} e^{-i\frac{q_x a}{2} - i\frac{q_y a}{2\sqrt{3}}}$$

$$+ e^{\pm i\frac{2\pi}{3}} e^{i\frac{q_x a}{2} - i\frac{q_y a}{2\sqrt{3}}} =$$

$$= e^{i\frac{q_y a}{\sqrt{3}}}\left(1 + e^{\mp i\frac{2\pi}{3}} e^{-i\frac{q_x a}{2} - i\frac{\sqrt{3}q_y a}{2}} + e^{\pm i\frac{2\pi}{3}} e^{i\frac{q_x a}{2} - i\frac{\sqrt{3}q_y a}{2}}\right)$$

$$\cancel{\frac{1}{\sqrt{3}}\left(1 + \frac{1}{2\sqrt{3}} + \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2}\right.}$$

$$= e^{i\frac{q_y a}{\sqrt{3}}}\left(1 + e^{\mp i\frac{2\pi}{3}}\left[1 - \frac{i}{2}(q_x a + \sqrt{3}q_y a)\right] + e^{\pm i\frac{2\pi}{3}}\left[1 - \frac{ia}{2}(-q_x + \sqrt{3}q_y)\right]\right)$$

$$= e^{i\frac{q_y a}{\sqrt{3}}}\left(1 + 2\cos\frac{2\pi}{3} - \frac{ia}{2}(q_x + \sqrt{3}q_y)e^{\mp i\frac{2\pi}{3}} - \frac{ia}{2}(-q_x + \sqrt{3}q_y)e^{\pm i\frac{2\pi}{3}}\right)$$

$$\xi(\pm\vec{K}+\vec{q}) = e^{i\frac{q_y a}{\sqrt{3}}}\left(-\frac{ia}{2}\right)\left\{q_x\left(e^{\mp i\frac{2\pi}{3}} - e^{\pm i\frac{2\pi}{3}}\right)\right.$$

$$\left. + \sqrt{3}\, q_y\left(e^{\mp i\frac{2\pi}{3}} + e^{\pm i\frac{2\pi}{3}}\right)\right\} =$$

$$= e^{i\frac{q_y a}{\sqrt{3}}}\left(-\frac{ia}{2}\right)\left(\mp 2i \underbrace{\sin\left(\frac{2\pi}{3}\right)}_{=\frac{\sqrt{3}}{2}} q_x + 2\underbrace{\cos\left(\frac{2\pi}{3}\right)}_{=-\frac{1}{2}} \sqrt{3}\, q_y\right)$$

$$= e^{i\frac{q_y a}{\sqrt{3}}}\left(-\frac{ia}{2}\right)\left(\mp i\sqrt{3}\, q_x - \sqrt{3}\, q_y\right) =$$

$$= e^{i\frac{q_y a}{\sqrt{3}}}\left(-\frac{\sqrt{3}a}{2}\right)\left(\pm q_x - i q_y\right) \cong \frac{\sqrt{3}ta}{2}\left(\pm q_x - i q_y\right)$$

insert $(-t)$ factor (p. 81)

Thus near point $\vec{K}$ (top sign in above equations)

$$\mathcal{H}_K(\vec{q}) = \begin{pmatrix} 0 & \hbar v(q_x - i q_y) \\ \hbar v(q_x + i q_y) & 0 \end{pmatrix} \qquad \boxed{v \equiv \frac{\sqrt{3}ta}{2\hbar}}$$

near $\vec{K}'$

$$\mathcal{H}_{K'}(\vec{q}) = \begin{pmatrix} 0 & \hbar v(-q_x - i q_y) \\ \hbar v(-q_x + i q_y) & 0 \end{pmatrix} = \begin{pmatrix} 0 & -v\hbar(q_x + i q_y) \\ -v\hbar(q_x - i q_y) & 0 \end{pmatrix}$$

let $\xi = \pm 1$ be <u>valley ~~isospin~~ index</u> (isospin)

$\xi = +1$ for $\vec{k} \approx \vec{K}$

$\xi = -1$ for $\vec{k} \approx -\vec{K} = \vec{K}'$.

$$\mathcal{H}_K(\vec{q}) = \mathcal{H}_{\xi=+1}(\vec{q}) = \hbar v\left(q_x \sigma^x + q_y \sigma^y\right)$$

$$\mathcal{H}_{K'}(\vec{q}) = \mathcal{H}_{\xi=-1}(\vec{q}) = -\hbar v\left(q_x \sigma^x - \xi q_y \sigma^y\right)$$

$$\sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\xi(\vec{k}) = (-t)\left[2\cos\left(\frac{k_x a}{2}\right) e^{-i\frac{k_y a}{2\sqrt{3}}} + e^{i\frac{k_y a}{\sqrt{3}}}\right]$$

$$\vec{K} = \frac{4\pi}{3a}\hat{x}$$

$$\vec{k} = \vec{K} + \vec{q} = \left(\frac{4\pi}{3a} + q_x, q_y\right)$$

MATH $$\xi_K(\vec{q}) = (-t)\left(-\frac{\sqrt{3}a}{2}\right)(q_x - iq_y)$$

now $\vec{k} = -\vec{K} + \vec{q} \Rightarrow \xi_{K'}(\vec{q}) = (-t)\left(\frac{\sqrt{3}a}{2}\right)(q_x + iq_y)$

$$\Rightarrow \mathcal{H}_K(\vec{q}) = \frac{\sqrt{3}ta}{2}\begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix}$$

$$\mathcal{H}_{K'}(\vec{q}) = -\frac{\sqrt{3}ta}{2}\begin{pmatrix} 0 & q_x + iq_y \\ q_x - iq_y & 0 \end{pmatrix}$$

Unified notation $\mathcal{H}_\xi(\vec{q}) = \xi \hbar v (q_x \sigma^x + \xi q_y \sigma^y)$

Eigenvalues (energy) are

$$\det \begin{pmatrix} -E & \xi \hbar v (q_x - i\xi q_y) \\ \xi \hbar v (q_x + i\xi q_y) & -E \end{pmatrix} = 0$$

$$\Rightarrow E^2 = (\xi \hbar v)^2 (q_x - i\xi q_y)(q_x + i\xi q_y) =$$

$$= (\hbar v)^2 (q_x^2 + q_y^2) = (\hbar v \vec{q})^2$$

$\Rightarrow E_\pm = \pm \hbar v |\vec{q}|$ (is independent of valley index $\xi = \pm 1$)

As before, let near $\vec{K}$ ($\xi = +1$)

$$\psi_+ = \begin{pmatrix} \psi_{A+} \\ \psi_{B+} \end{pmatrix} \text{ so that } \hbar v \begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix} \begin{pmatrix} \psi_{A+} \\ \psi_{B+} \end{pmatrix} = E \begin{pmatrix} \psi_{A+} \\ \psi_{B+} \end{pmatrix}$$

But near $\vec{K}' = -\vec{K}$ we choose (interchange $q_x + iq_y$, $q_x - iq_y$ elements)

$$\psi_- = \begin{pmatrix} \psi_{B-} \\ \psi_{A-} \end{pmatrix} \text{ so that } \cancel{-\hbar v \begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix} \begin{pmatrix} \psi_{B-} \\ \psi_{A-} \end{pmatrix}}$$

$$-\hbar v \begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix} \begin{pmatrix} \psi_{B-} \\ \psi_{A-} \end{pmatrix} = E \begin{pmatrix} \psi_{B-} \\ \psi_{A-} \end{pmatrix}$$

(That is, for $\vec{k}$ near $\vec{K}'$ we interchange sublattices A and B)

Then near both points $\mathcal{H}$ can be written as

$$\mathcal{H} = \hbar v \, \tau^z \otimes \vec{\sigma} \cdot \vec{q} = \hbar v \begin{pmatrix} \vec{\sigma}\cdot\vec{q} & 0 \\ 0 & -\vec{\sigma}\cdot\vec{q} \end{pmatrix} =$$

wave function

$$\psi_K = \begin{pmatrix} \psi_{A+} \\ \psi_{B+} \\ \psi_{B-} \\ \psi_{A-} \end{pmatrix} \qquad = \hbar v \begin{pmatrix} 0 & q_x - i q_y & 0 & 0 \\ q_x + i q_y & 0 & 0 & 0 \\ 0 & 0 & 0 & -(q_x - i q_y) \\ 0 & 0 & -(q_x + i q_y) & 0 \end{pmatrix}$$

$$\tau^z \otimes \vec{\sigma} = \begin{pmatrix} \vec{\sigma} & 0 \\ 0 & -\vec{\sigma} \end{pmatrix} \qquad \left(\begin{array}{l} \vec{\tau} \text{ acts in valley space} \\ \vec{\sigma} \text{ acts in sublattice A/B space} \end{array} \right)$$

$$\vec{\sigma}\cdot\vec{q} = \sigma^x q_x + \sigma^y q_y = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} q_x + \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} q_y =$$

$$= \begin{pmatrix} 0 & q_x - i q_y \\ q_x + i q_y & 0 \end{pmatrix}$$

$$\hat{\mathcal{H}} \psi = E \psi$$

look just like 4-dimensional relativistic Dirac equation!

Near $\xi = +1$: $\hbar v \begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix} \begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix} = \lambda \hbar v |\vec{q}| \begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix}$

$$(q_x - iq_y)\psi_B = \lambda |\vec{q}| \psi_A$$

$$\psi_A = \lambda \frac{q_x - iq_y}{\sqrt{q_x^2 + q_y^2}} \psi_B = \lambda \frac{|\vec{q}| e^{-i\varphi_q}}{|\vec{q}|} \psi_B$$

$$e^{-i\varphi_q} \equiv \cos\varphi_q - i \sin\varphi_q$$

$$\left.\begin{aligned} |\vec{q}| \cos\varphi_q &= q_x \\ |\vec{q}| \sin\varphi_q &= q_y \end{aligned}\right\} \Rightarrow \tan\varphi_q = \frac{q_y}{q_x}$$

$$\varphi_q = \tan^{-1}\left(\frac{q_y}{q_x}\right)$$

$$\Rightarrow \psi_{\xi=+1} = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{-i\varphi_q} \\ \lambda \end{pmatrix} = \frac{e^{-i\varphi_q}}{\sqrt{2}} \begin{pmatrix} 1 \\ \lambda e^{-i\varphi_q} \end{pmatrix}.$$